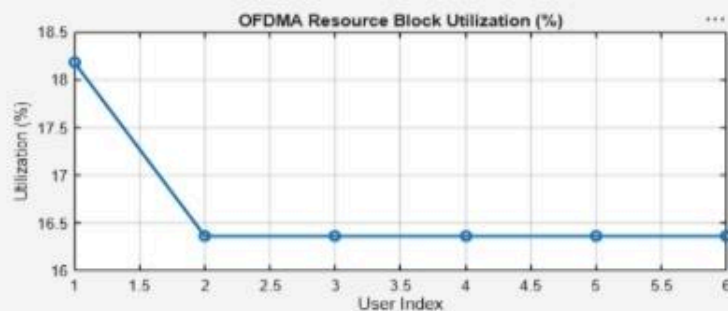
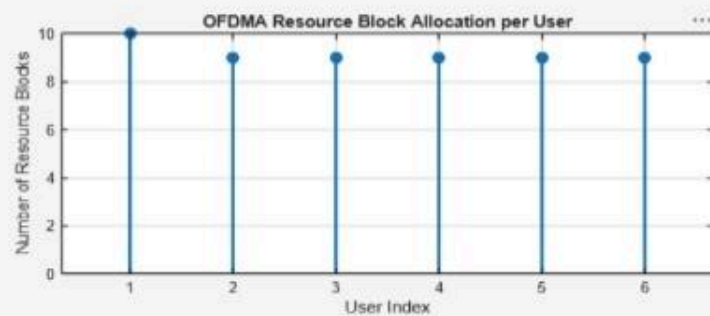


Ex.No.4: OFDMA Resource Block Allocation

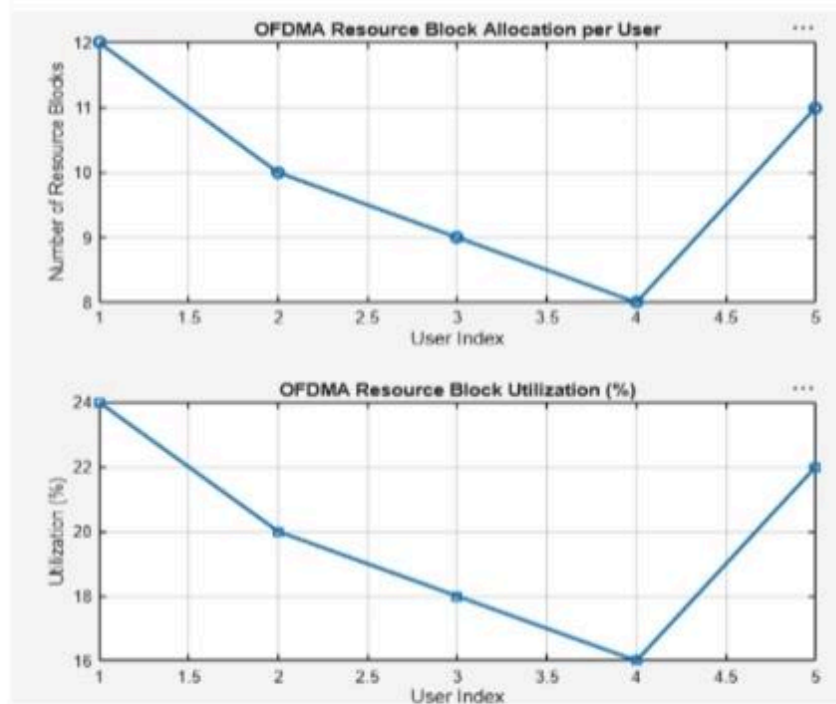
Objective 1: Matlab Code and Output

```
%MATLAB SCRIPT FOR OFDMA RB ALLOCATION
1  clc; clear; close all;
2  % System parameters
3  bandwidth = 10e6;
4  rb_size = 180e3;
5  % Total Resource Blocks
6  % 10 MHz
7  % 180 kHz per RB
8  total_RB = floor(bandwidth / rb_size);
9  % Users
10 users = 1:6;
11 % RB allocation (simple equal + remainder distribution)
12 base = floor(total_RB / length(users));
13 alloc = base * ones(1,length(users));
14 alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
15 utilization = (alloc / total_RB) * 100;
16 disp('Total Resource Blocks:');
17 disp(total_RB)
18 disp('RB Allocation per User:');
19 disp(alloc)
20 figure
21 % ----- Graph 1: RB Allocation per User -----
22 subplot(2,1,1)
23 stem(users, alloc, 'filled', 'LineWidth', 2)
24 title('OFDMA Resource Block Allocation per User')
25 xlabel('User Index')
26 ylabel('Number of Resource Blocks')
27 grid on
28 % ----- Graph 2: RB Utilization -----
29 subplot(2,1,2)
30 plot(users, utilization, '-o', 'LineWidth', 2)
31 title('OFDMA Resource Block Utilization (%)')
32 xlabel('User Index')
33 ylabel('Utilization (%)')
34 grid on
```



Objective 2: Matlab Code and Output

```
36 % Total system resource blocks
37 total_RB = 50;
38 % Number of active users
39 users = 1:5;
40 % RB allocation to users (example scenario)
41 alloc = [12 10 9 8 11];
42 % Resource Block Utilization (%)
43 util = (alloc / total_RB) * 100;
44 disp('Resource Block Utilization (%) per User:')
45 disp(util')
46 figure
47 % ----- Graph 1: RB Allocation -----
48 subplot(2,1,1)
49 plot(users, alloc, '-o', 'LineWidth', 2)
50 title('OFDMA Resource Block Allocation per User')
51 xlabel('User Index')
52 ylabel('Number of Resource Blocks')
53 grid on
54 % ----- Graph 2: Utilization (%) -----
55 subplot(2,1,2)
56 plot(users, util, '-s', 'LineWidth', 2)
57 title('OFDMA Resource Block Utilization (%)')
58 xlabel('User Index')
59 ylabel('Utilization (%)')
60 grid on
61
```



Objective 3: Matlab Code and Output

```
62 % System parameters
63 rb_bw = 180e3;
64 users = 1:5;
65 % Bandwidth per RB (180 kHz)
66 % Allocated Resource Blocks per user
67 alloc_RB = [12 10 9 8 11];
68 % Modulation efficiency (bits/symbol)
69 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
70 % Throughput calculation (Mbps)
71 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
72 disp('User Throughput (Mbps):')
73 disp(throughput)
74 figure
75 % ----- Graph 1: Throughput per User -----
76 subplot(2,1,1)
77 plot(users, throughput, '-o', 'LineWidth', 2)
78 title('OFDMA User Throughput')
79 xlabel('User Index')
80 ylabel('Throughput (Mbps)')
81 grid on
82 % ----- Graph 2: RB vs Throughput -----
83 subplot(2,1,2)
84 plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
85 title('Resource Blocks vs Throughput Relationship')
86 xlabel('Allocated Resource Blocks')
87 ylabel('Throughput (Mbps)')
88 grid on
```

